

EthosAgentAI: An Ethical, Multi-Model Agentic AI Ecosystem for Fair Hiring and Intelligent Skill Development Aligned with SDG 8 and SDG 9

Presented by

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Outline

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Introduction

Every year, thousands of Bangladeshi graduates enter the job market with hope but face unclear job requirements, biased screening, difficult online interviews and Personalized coach. Recruiters also struggle with huge applicant volumes and rising interview fraud. Seeing these real challenges inside DIU and my office, I asked: “Can we build an AI system that helps both students and recruiters fairly, transparently, and intelligently?” Under the supervision of Dr. S. M. Aminul Haque, I developed EthosAgentAI end-to-end AI ecosystem combining job matching, resume screening, personalised roadmaps, and real-time interview integrity. Now, it is ready to be integrated into the Daffodil Career Hub to support fresh graduates as well as industry.

Objective

- To design and develop a unified multi-agent AI framework capable of supporting end-to-end hiring and personalized career intelligent workflows.
- To implement semantic job-matching mechanisms with explainable and transparent decision-making.
- To develop an intelligent resume-screening module utilizing RAG, adaptive scoring, and structured evaluation criteria.
- To identify candidate skill gaps and generate personalised, data-driven learning roadmaps.
- To implement real-time interview integrity monitoring using MediaPipe, OpenCV, OCR, and vision-based detection models.
- To integrate a personalised AI career-assistant chatbot for continuous guidance and interaction.
- To achieve robust system performance, targeting $\geq 85\%$ accuracy in resume screening and 95–100% accuracy in interview-integrity detection.
- To ensure that the system aligns with SDG 8 (Decent Work and Economic Growth) and SDG 9 (Industry, Innovation and Infrastructure), supporting sustainable and ethical recruitment practices.

Background Study

- Recruitment processes are increasingly digital, yet face challenges such as bias, inefficiency, and online interview fraud.
- Job seekers often struggle with skill gaps and lack guidance to improve employability.
- Recruiters face high applicant volumes, limited time, and manual screening constraints.
- Existing systems are often fragmented: some focus only on job matching, others on resume screening, skill-gap analysis, or interview monitoring.
- Research shows AI-based solutions (semantic matching, multi-agent screening, RAG, computer vision) can improve fairness, efficiency, and transparency.
- In the context of Bangladesh, fresh graduates need personalised guidance aligned with industry expectations, while organizations require scalable and reliable evaluation systems.

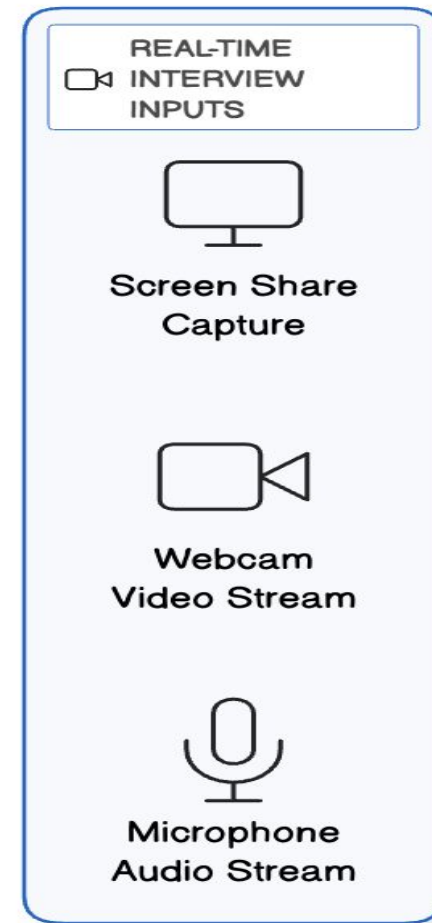
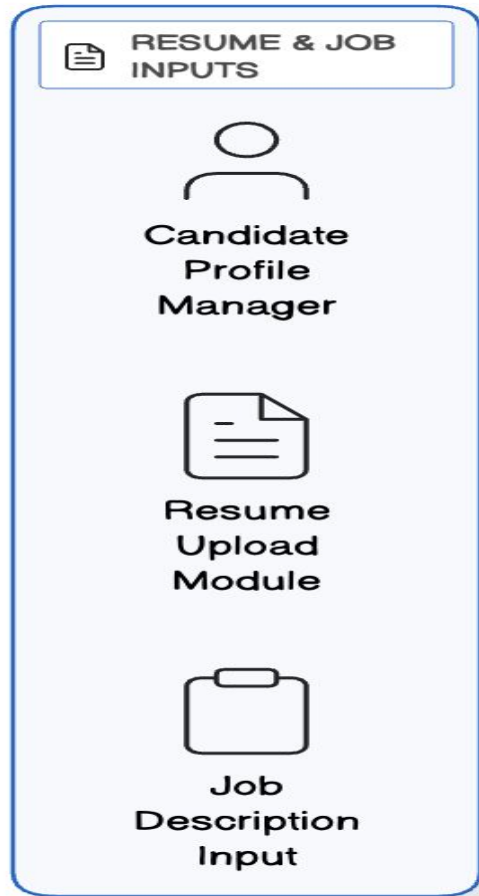
This gap motivates the development of EthosAgentAI, a unified, end-to-end AI recruitment and skill-development ecosystem.

Gap Analysis

Current Solutions	Identified Gaps	EthosAgentAI Innovation
AI tools (HireVue, Eightfold, XOPA) handle isolated tasks like screening or interviews	No end-to-end recruitment ecosystem	✓ Unified 5-layer architecture (Input → Processing → Knowledge → Decision → Output)
Academic research focuses on single-phase automation	Lack of full hiring lifecycle automation	✓ Complete recruitment-to-career pipeline
Black-box AI decision making	Weak explainability & transparency	✓ Explainable AI (XAI) & interpretable decisions
Biased datasets and biased decision risks	No built-in fairness auditing	✓ Fairness-by-design & bias mitigation framework
No real-time interview monitoring	Interview integrity not ensured	✓ Hybrid computer vision for real-time fraud detection
Recruitment systems disconnected from learning platforms	No skill-gap to learning roadmap link	✓ Personalized skill-gap analysis & adaptive learning path
Single-agent or rule-based systems	No multi-agent reasoning ecosystem	✓ Multi-agent system with RAG-enhanced intelligence
Closed, proprietary platforms	No open or reproducible research system	✓ Open, ethical, and SDG-aligned AI framework

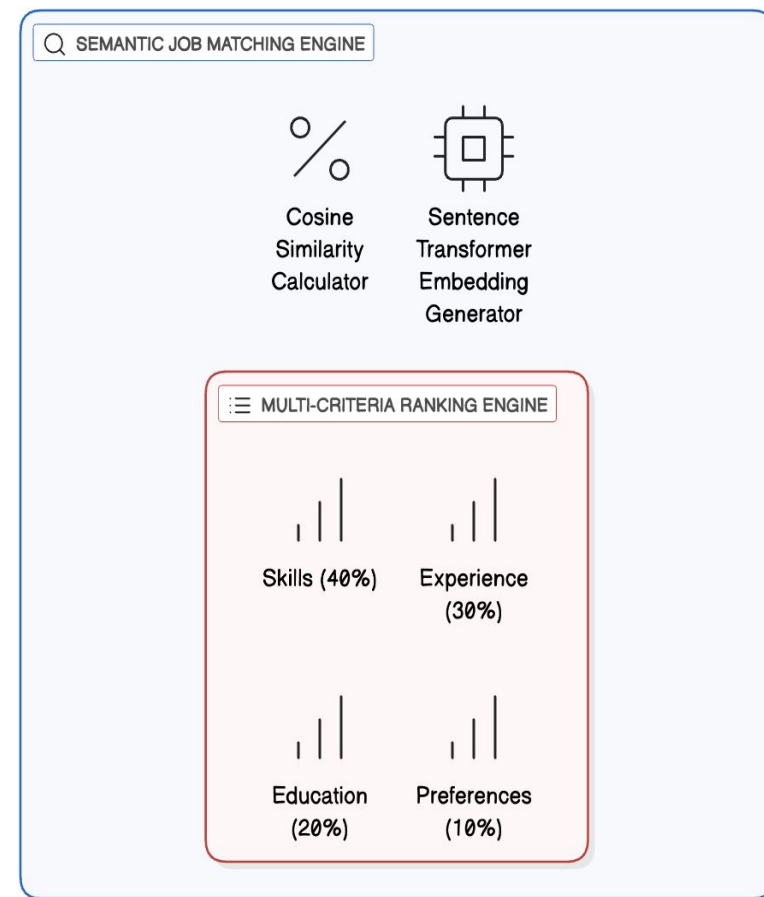
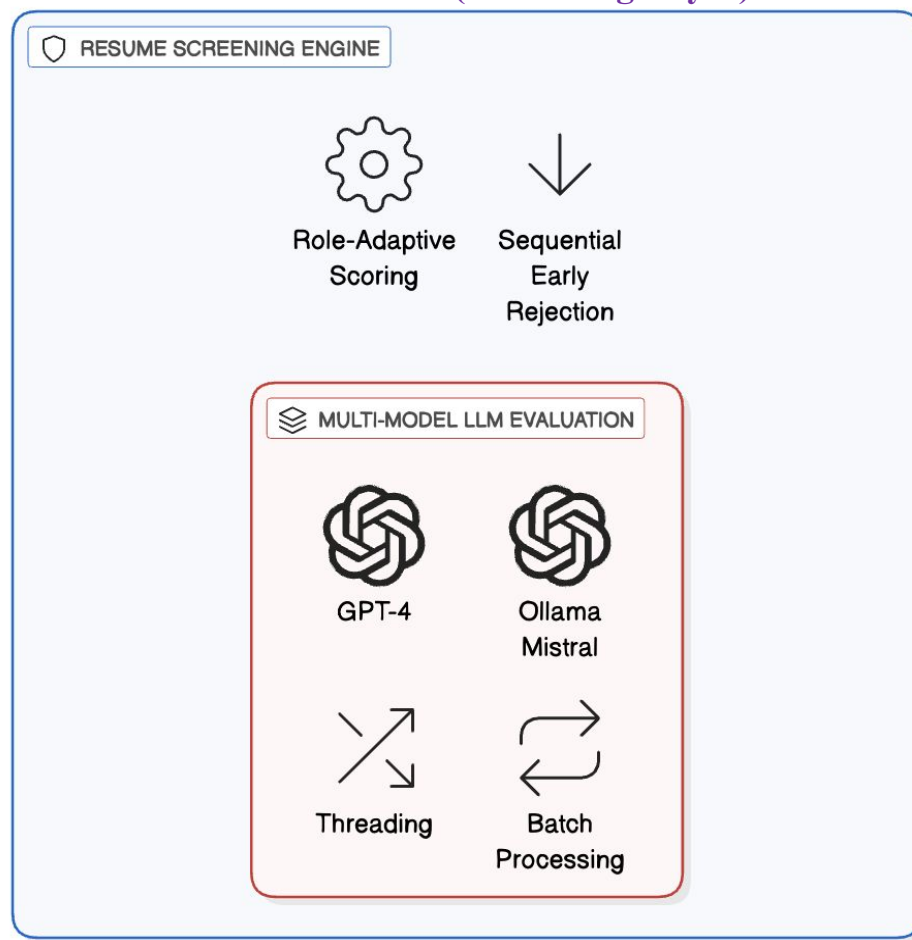
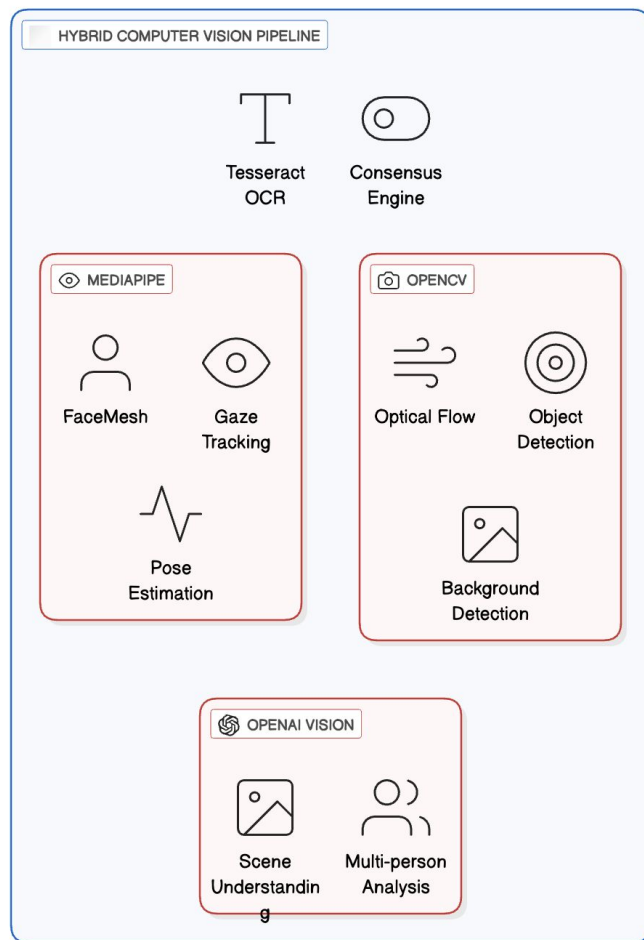
Methodology

(Input Layer)



Methodology

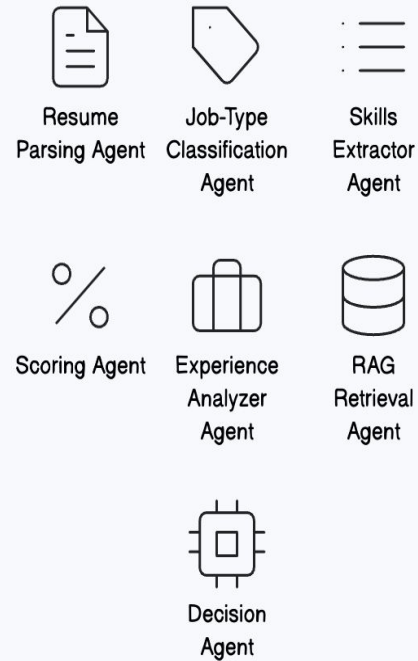
(Processing Layer)



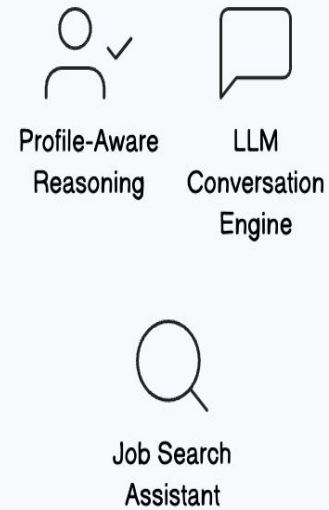
Methodology

(Processing Layer)

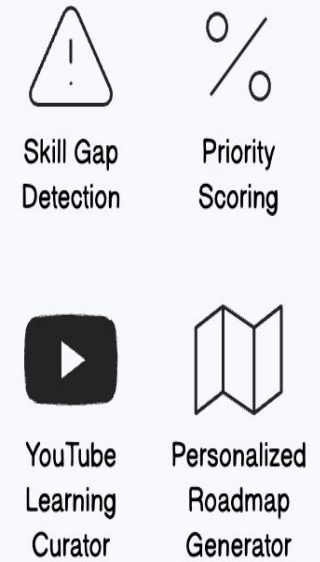
MULTI-AGENT ORCHESTRATION



AI CAREER CHATBOT

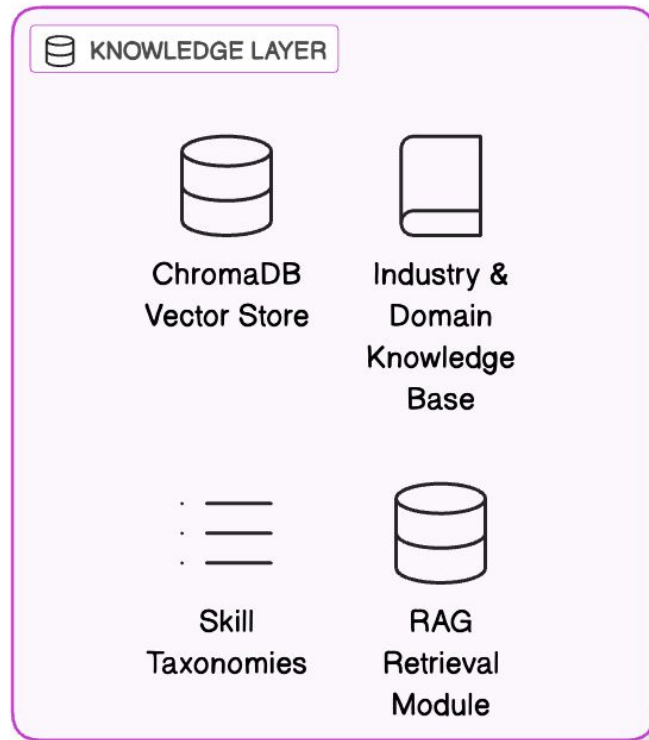


SKILL-GAP ANALYZER & LEARNING ROADMAP



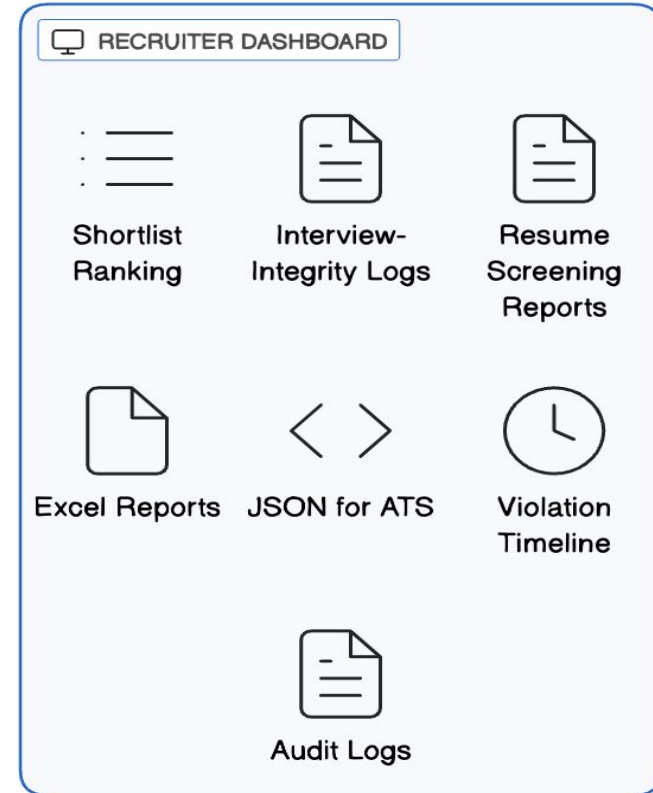
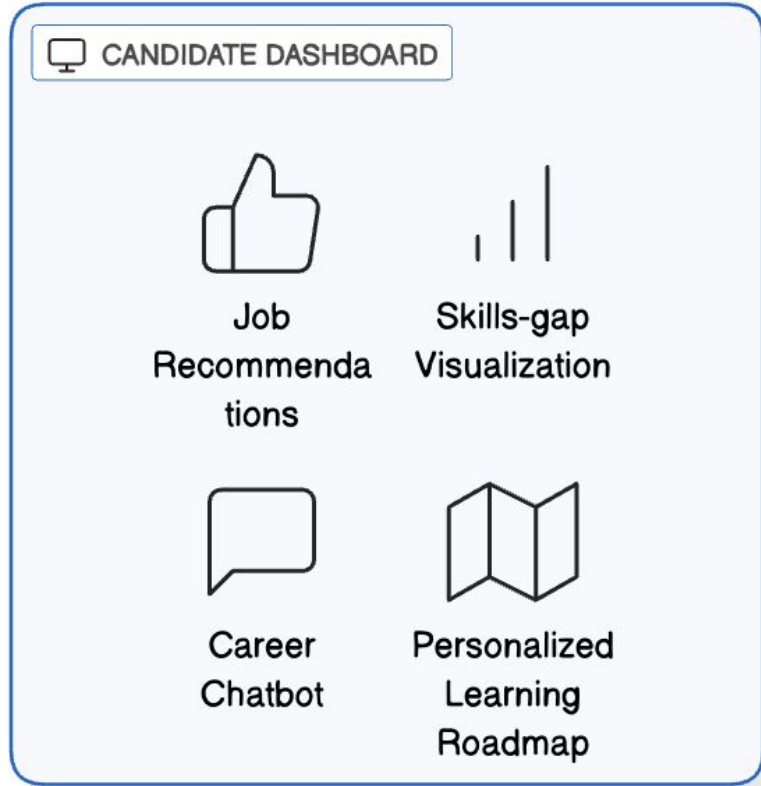
Methodology

(Knowledge and Decision Layer)



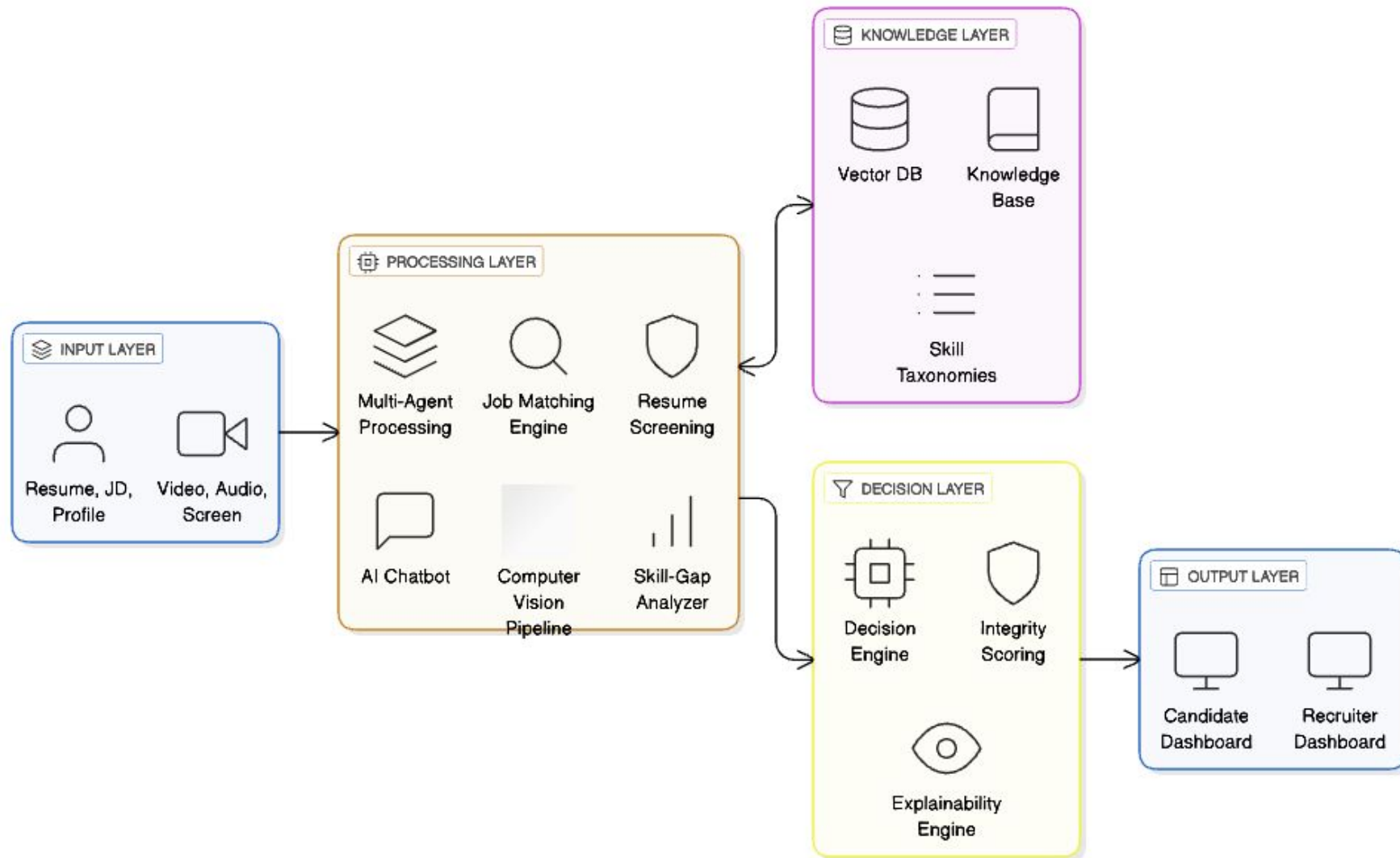
Methodology

(Output Layer)



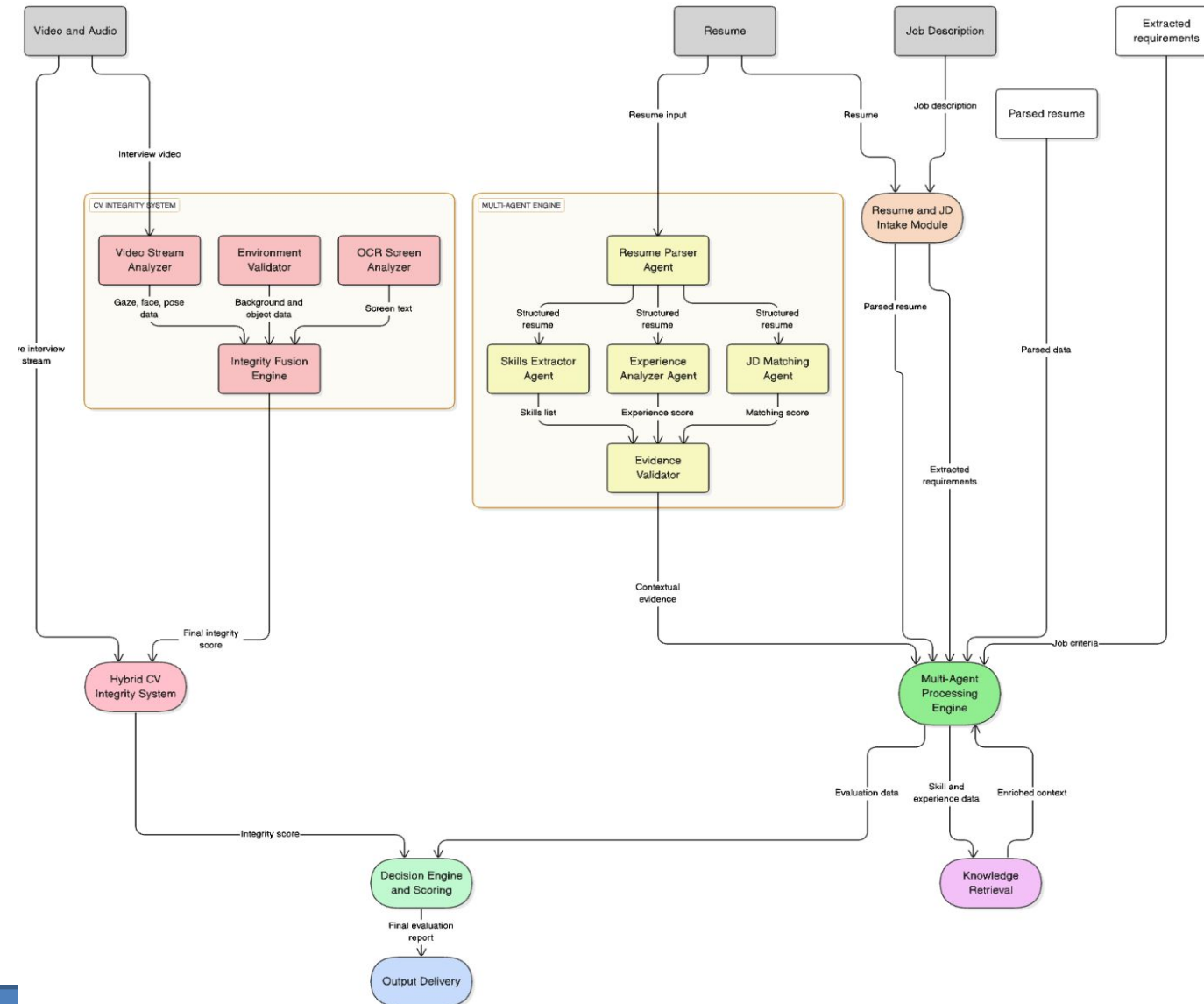
Methodology

(System Architecture Overview)



Methodology

(Data Flow Overview)



Result and Analysis

Resume Screening Results

- Accuracy: 86% (vs Keyword 62%, Single-LLM 78%),
- Precision: 0.84,
- Recall: 0.86,
- F1-Score: 0.85

System Efficiency

- Avg. Screening Time: 22.6 sec / resume (60% faster than manual),
- Early Rejection Processing: 18–22 sec,
- Interview Latency: <100 ms,
- Dashboard Response: 1.8–2.1 sec

Interview Integrity Accuracy

- MediaPipe (Face & Pose): 96% ,
- OpenCV Motion Detection: 93%,
- OCR Content Detection: 92% ,
- Vision Model (Scene Analysis): 97%,
- Consensus Accuracy: 95–100%,
- False Positives: <5%

Job Matching Results

- Semantic Accuracy: 88% (vs Keyword 61%, Embedding-only 77%)

Analysis

- Results confirm that EthosAgentAI significantly outperforms traditional keyword-based and single-model systems in both screening and matching accuracy.
- High interview integrity accuracy with low false positives proves the reliability of the hybrid computer vision framework.
- System efficiency improvements and low latency demonstrate practical real-world deployment feasibility.
- Additionally, the platform supports personalized learning paths, enabling continuous skill development and adaptive feedback for candidates, further enhancing engagement and long-term growth.
- Higher consistency, explainability, and bias resistance than human recruitment establish EthosAgentAI as a trustworthy ethical hiring system.

Novelty of the Work

- Architectural: Integrates screening, matching, skills assessment, integrity checks, and guidance in a five-layer pipeline with hybrid cloud-local fallback.
- Algorithmic: Consensus-based CV, role-adaptive scoring, sequential rejection for efficiency.
- Methodological: First use of RAG in recruitment, LangGraph multi-agent coordination, explanation-first assessment.
- Social/Ethical: Fairness-by-design, dual-sided support (candidates/recruiters), direct SDG 8 & 9 alignment.
- Advances State-of-the-Art: Overcomes fragmentation in existing tools; Medium familiarity blending established (NLP, APIs) and novel (end-to-end fusion, behavioral consensus) elements.
- Impact: Reduces bias, enhances reproducibility; Promising for future responsible AI systems.

Conclusion

EthosAgentAI is a personalized, multi-agent AI ecosystem that enhances hiring efficiency, equity, and transparency using advanced NLP, RAG, and hybrid computer vision. It achieves 86% resume screening accuracy, 95–100% interview integrity, and 88% semantic job matching. The platform uniquely delivers personalized skill-gap analysis, curated learning roadmaps, and an intelligent career-assistant chatbot, empowering candidates end to end. It advances SDG 8 by democratizing fair hiring and personalized career support, and supports SDG 9 through scalable, ethical AI innovation shifting recruitment from automation to lifelong personalized career intelligence.

Future Work:

Future enhancements include multimodal evaluation (text, visual, behavioral), predictive career trajectory modeling, and stronger bias mitigation with responsible AI alignment. Integration into the Daffodil Career Hub and nationwide rollout will enable large-scale, ethical deployment, making hiring and career development smarter, fairer, and more personalized.

References

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THANK YOU